

Industrial Water Pump Maintenance Manual (Sample)

Overview

This sample manual describes routine maintenance procedures for an industrial centrifugal water pump used in manufacturing plants.

Safety

Disconnect electrical power before servicing. Wear gloves, safety glasses, and lock out the machine before inspection.

Daily Inspection

Check for unusual noise, vibration, oil leakage, overheating, and coolant level.

Weekly Maintenance

Inspect bearings, tighten bolts, verify motor alignment, and clean air vents.

Monthly Maintenance

Lubricate bearings, inspect seals, examine the impeller for wear, and clean the suction filter.

Error Code E101

Cause: Low coolant level. Action: Refill coolant and inspect for leaks.

Error Code E202

Cause: Bearing overheating. Action: Stop the pump, inspect lubrication, replace damaged bearings if required.

Error Code E303

Cause: Excessive vibration. Action: Check shaft alignment, tighten mounting bolts, and inspect the coupling.

Error Code E404

Cause: Motor overheating due to a blocked cooling fan or excessive load. Corrective actions: Shut down the pump, allow the motor to cool, clean the cooling fan, inspect motor current, replace the fan if damaged, and restart while monitoring temperature.

Recommended Spare Parts

Cooling fan assembly, shaft seal, bearings, impeller, coupling, gasket kit.

Preventive Maintenance

Perform daily inspection, weekly cleaning, monthly lubrication, quarterly alignment checks, and annual bearing inspection.

Troubleshooting

If the pump fails to start, verify the power supply, inspect the motor protection relay, and check the control panel for fault indicators.